# Predicting Falcon 9 First-Stage Landing Success

Analyzing historical SpaceX launch data to develop predictive machine learning models for booster recovery outcomes and launch cost optimization.

**DATA SCIENCE CAPSTONE PROJECT | IBM SKILLS NETWORK**

# Executive Summary

| 90 | 66.7% | 83.3% | 0.889 |
|---|---|---|---|
| TOTAL LAUNCHES | LANDING LABEL RATE | BEST TEST ACCURACY | PEAK F1-SCORE |

This project analyzes 90 historical Falcon 9 launches (2010–2020) to predict first-stage recovery success. Using a tuned **Random Forest** classifier, we achieved 83.3% test accuracy. Key findings indicate that technical features—specifically the presence of **Grid Fins** and **Legs**—are the strongest predictors of a successful landing outcome.

SOURCE: IBM CAPSTONE CURATED DATASET (N=90)

# Introduction: Problem, Objective, and Target

## WHY PREDICT RECOVERY?

SpaceX advertises Falcon 9 launches at roughly $62 million, versus more than $165 million for other providers in the historical capstone case. Reusing the first stage is a major source of this cost advantage. Predicting recovery outcomes helps a competitor estimate launch economics using public mission information.

## RESEARCH QUESTIONS

- How do orbit, payload mass, launch site, and booster reuse relate to the landing label?
- How did the annual landing-label rate change in the 2010–2020 sample?
- Which binary classifier performs best under a fixed holdout evaluation?

## TARGET DEFINITION

**CLASS = 1**

The curated dataset marks a successful first-stage landing, including RTLS and drone-ship recovery outcomes.

**CLASS = 0**

All other coded outcomes, including unsuccessful recovery and some missions with no landing attempt or unspecified result. Thus this label rate is not conditional success among attempted landings.

# Data Collection: REST API and Web Scraping

## 1. SPACEX REST API V4

Collected launch, rocket, payload, and launch-site fields. The direct endpoint was unavailable during this run; a frozen IBM Skills Network response snapshot was used instead.

```
GET https://api.spacexdata.com/v4/launches/past
```

## 2. WIKIPEDIA TABLE SCRAPING

Downloaded an archived launch-list revision, parsed its HTML tables with pandas, retained dated rows from 2010–2020, and preserved the Falcon 9/Heavy scrape separately from the labeled Falcon 9 sample.

```
pandas.read_html(archived_wikipedia_html)
```

PUBLIC NOTEBOOK REPOSITORY (REFERENCE):
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone

## DATA INVENTORY AND LAB LINKS

| Artifact | Coverage | Rows |
|---|---|---|
| IBM API snapshot | Historical | 107 |
| Curated Falcon 9 modeling table | 2010–2020 | 90 |
| Archived Wikipedia scrape | 2010–2020 | 114 |

DIRECT CAPSTONE LAB NOTEBOOKS:
Data Collection API.ipynb
Data Collection with Web Scraping.ipynb
Data Wrangling.ipynb
EDA with Data Visualization.ipynb
EDA with SQL.ipynb
Interactive Visual Analytics with Folium.ipynb
Machine Learning Prediction.ipynb

# Data Wrangling and Feature Engineering

## LABELED DATASET OVERVIEW (N=90)

| ID | DATE | PAYLOAD (KG) | LAUNCH SITE | ORBIT | CLASS |
|---|---|---|---|---|---|
| 01 | 2010-06-04 | 6104.9 | CCAFS SLC 40 | LEO | 0 |
| 56 | 2018-08-07 | 5800.0 | CCAFS SLC 40 | GTO | 1 |
| 78 | 2020-05-30 | 9525.0 | KSC LC 39A | ISS | 1 |
| 90 | 2020-11-05 | 3681.0 | CCAFS SLC 40 | MEO | 1 |

**Cleaning Note:** Excluded LandingPad due to 26 missing values (28.8% sparsity) to prevent training bias and potential target leakage.

## PROCESSING WORKFLOW

> **Filtering:** Retained Falcon 9 launches; excluded experimental and Falcon Heavy flights.

> **Labeling:** Converted qualitative outcomes into binary target: 1 (Success) vs. 0 (Other).

> **Selected Predictors:**

`PayloadMass` `Flights` `Block` `Orbit` `LaunchSite` `GridFins` `Legs`

> **Preprocessing:** One-Hot Encoding for categorical; StandardScaler for numerical features.

# EDA and Data Visualization Methodology

## SQL AGGREGATION

```
SELECT LaunchSite, COUNT(*) AS n,
AVG("Class") * 100 AS rate
FROM spacex
GROUP BY LaunchSite;
```

SQLite queries aggregate launch outcomes by site, year, orbit, and payload class. Counts are retained beside every percentage to make denominators explicit and avoid over-interpreting rare groups.

Visualizations use horizontal rate bars and annual trends; mean payloads are compared by class. A separate model pipeline performs encoding/scaling only after the train/test split.

## OBSERVED SITE RESULTS · N=90

| Launch Site | N | Class 1 | Rate |
|---|---|---|---|
| KSC LC 39A | 22 | 17 | **77.3%** |
| VAFB SLC 4E | 13 | 10 | **76.9%** |
| CCAFS SLC 40 | 55 | 33 | **60.0%** |

Kennedy LC 39A leads by observed rate, but CCAFS supplies 61% of the sample (55/90). Differences are descriptive; the sites serve different mission mixes.

# Annual Landing Success Rate



CLASS 1 SHARE BY YEAR · BAR LENGTH = SUCCESS RATE · N = LAUNCHES IN YEAR




**0% → 90%**
Share of positive landing labels rose between the first modeled launch (2010) and 2019.

**2017: N=18**
The rate varies across years; 2018 falls to 61.1% before rising again.

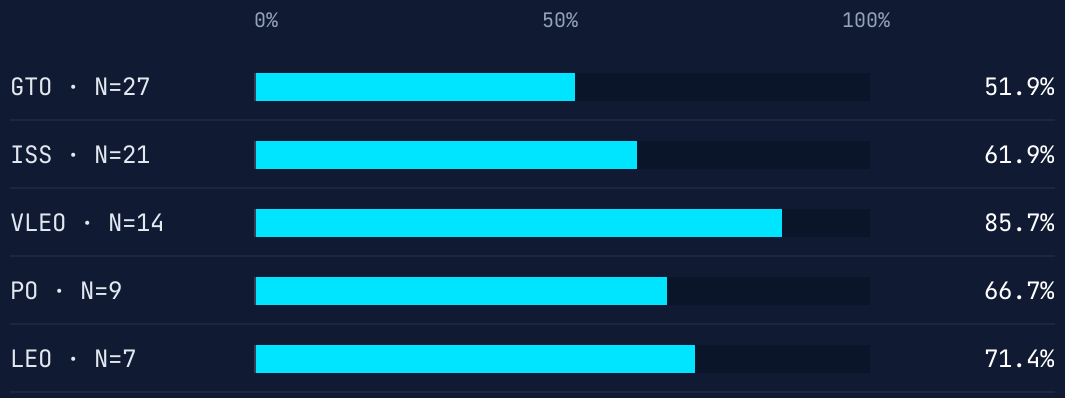
**Interpret carefully**
Early-year denominators are tiny; Class 0 also includes missions without an attempt.

SOURCE: IBM CURATED FALCON 9 DATASET · GROUPED BY YEAR

# Site, Orbit, and Payload EDA

## LANDING-LABEL RATE BY ORBIT

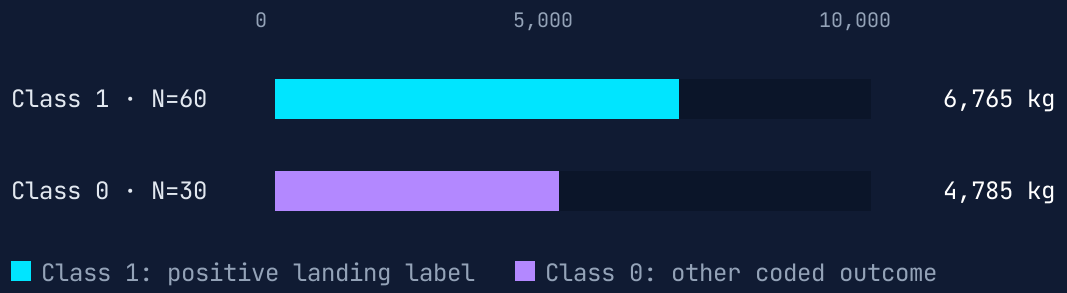
SUCCESS RATE (%) · DENOMINATOR SHOWN AS N




> GTO has the largest orbit sample (N=27) and the lowest rate among the five most common orbit groups; the observed difference is descriptive, not causal.

## MEAN PAYLOAD MASS BY CLASS

PAYLOAD MASS (KG) · BAR SCALE 0–10,000 KG




> The positive-label group has a higher mean payload in this sample. This is an association; orbital energy and recovery profile also affect landing feasibility.

Note: some orbit rates at 100% are based on only one record. See the accompanying CSV outputs for all orbit categories and exact counts.

SOURCE: IBM CURATED DATASET · ORBIT AND PAYLOAD GROUPINGS

# Interactive Analytics: Folium Map and Plotly Dash

## FOLIUM · LAUNCH RECORDS & PROXIMITY




• site summary · • other site · site distances geodesic

| | |
|---|---|
| KSC LC 39A ↔ CCAFS SLC 40 | 5.8 km |
| VAFB SLC 4E ↔ nearest Florida site | 3,821.6 km |
| Map records | 90 launches · clustered markers |

Each launch popup includes flight number, date, site, payload, orbit, and Class. The interactive map includes separate record-cluster and site-summary layers. Proximity is calculated between launch sites using haversine distance—not to coastline or rail infrastructure.

## PLOTLY DASH · INTERACTIVE ANALYSIS




**Implemented in dashboard.py:**

- Site dropdown filters records and recalculates launch count and label rate.
- Plotly line chart shows yearly positive-label rate for the selected site.
- Scatter plot compares payload mass and flight number; hover shows date, site, orbit, and outcome.

FILES IN SUBMISSION: folium_launch_sites.html · dashboard.py

# Modeling Method and Comparison

## MACHINE LEARNING PIPELINE

01 **Splitting:** Stratified 80/20 train-test split (N_train=72, N_test=18).

02 **Validation:** 5-fold Stratified CV for optimal hyperparameter tuning.

03 **Optimization:** GridSearch over C, kernel, depth, and neighbors.

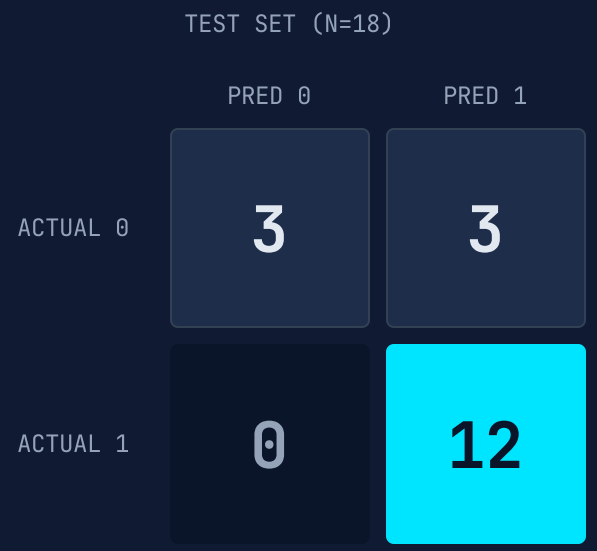
## CLASSIFIERS EVALUATED

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors
- Random Forest

**Metric Selection:** F1-Score and ROC-AUC were prioritized alongside Accuracy to account for class imbalance (66.7% success rate).

# Best Model and Test Results

## CONFUSION MATRIX: RANDOM FOREST




Model achieved perfect recall for landing labels (1.0), but incorrectly predicted 3 negative cases as positive.

## HOLDOUT SET PERFORMANCE

| METRIC | RESULT |
|---|---|
| Test Accuracy | 83.3% |
| Recall (Sensitivity) | 100.0% |
| Precision | 80.0% |
| F1-Score | 0.889 |
| ROC-AUC Score | 0.875 |

> **Analysis:** Random Forest provided the most robust fit. Due to the small N_test=18, individual classification errors significantly impact precision; larger datasets are needed for deployment.

RESULTS: RANDOM FOREST CLASSIFIER (N_TEST=18)

# Conclusion, Limitations, and References

## KEY FINDINGS

- Random Forest achieved 83.3% test accuracy (N_test=18); it detected all 12 positive test cases but produced three false positives.
- The annual positive-label rate rose from 0% in early years to 84.2% in 2020; yearly sample sizes vary from N=1 to N=19.
- KSC LC 39A has the highest observed site rate (77.3%, N=22); CCAFS SLC 40 has 60.0% (N=55). Site and mission mix are confounded.
- Mean payload is 6,765 kg for Class 1 vs. 4,785 kg for Class 0. This is an association, not a causal effect.
- KSC LC 39A and CCAFS SLC 40 are approximately 5.8 km apart; distances are great-circle calculations between pads.

**Limitations:** N=90, only 18 test cases, historical cut-off in November 2020, and Class 0 includes some missions without a landing attempt. Use a larger, recent, attempt-only sample before operational deployment. This analysis establishes neither causality nor a direct launch-cost estimate.

## REPOSITORY AND DIRECT LAB LINKS

**Public reference repository (full URL):**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone

**Data Collection API:**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone/blob/main/Data%20Collection%20API.ipynb

**Web Scraping:**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone/blob/main/Data%20Collection%20with%20Web%20Scraping.ipynb

**Data Wrangling:**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone/blob/main/Data%20Wrangling.ipynb

**EDA Visualization:**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone/blob/main/EDA%20with%20Data%20Visualization.ipynb

**EDA SQL:**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone/blob/main/EDA%20with%20SQL.ipynb

**Folium Visual Analytics:**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone/blob/main/Interactive%20Visual%20Analytics%20with%20Folium.ipynb

**Machine Learning:**
https://github.com/evgenyzorin/IBM-Applied-Data-Science-Capstone/blob/main/Machine%20Learning%20Prediction.ipynb

END OF SEQUENCE